Week 10: Concept Test Plan

1. What is the purpose of the concept test?
    a. What we are looking to determine is if senior citizens would be able to utilize SpinShelf, a stand-alone cabinet at waist-level where pushing an identified button causes it to rotate and come forward. What we are looking to understand is:
        - it's easier than how they get items from cabinets now
        - they'd trust a motorized cabinet
        - the buttons, height, size and rotation speed work for them
        - they'd want one in their home
2. Who are the target users invited to this test?
    a. Adults over the age of 65 living at home, preferably alone, and who frequently access objects from kitchen or pantry cabinets. Our target is a total of 5–6 users, such as:
        i. individuals who have difficulty accessing high or low cabinets
        ii. individuals who are technologically savvy and others who aren't
3. What is/are testing format?
    a. Face-to-face sessions, one user at a time, approximately 20 minutes long. Sessions will be divided into three segments:
        i. Introduction: We will present the idea using drawings and storyboarding.
        ii. Prototype testing: The user will operate the cardboard prototype. It will be done "Wizard of Oz" fashion. Once they press the button, the teammate will turn the shelf manually to mimic the motor.
        iii. Rating scale questionnaire and interviews: Rating scale questions and follow-up interviews.
4. How will you communicate the concept with the participants? (Please list out different ways and materials that your team will be used during the test. Please add images or reference of the materials
    a. Concept sketches: the appearance of SpinShelf and its placement in a kitchen setting.
    b. Storyboard: an elderly person moves from being unable to access a high cabinet to simply pressing the button of SpinShelf to retrieve their item. [insert image]
    c. Cardboard prototype: a cardboard tube divided into sections, placed on a lazy susan mechanism and elevated to the level of the waist with the help of a stand. The device has large buttons labeled one for each section and features actual household objects, such as a mug, some spices and a cereal box. [insert photo]
    d. Explanation and demonstration: the interviewer performs one complete button operation prior to the participant using it.
5. What is the content of the concept test (Please list out the introduction script and all questions. Refer to the "example of concept test survey of the electric scooter" in the lecture note)

    a. ***Introduction script:***

        i. Hi there, thanks for coming to meet us. We are UC Berkeley students who have developed a product called SpinShelf. Many accidents that occur at home occur

as a result of people reaching up to high shelves or bending down to reach low shelves. SpinShelf is basically a cabinet which is mounted at a height that is around waist level. Each compartment in the cabinet has a button; when you press the button, the whole compartment rotates to face you. That way, you can get what you want without having to either bend down or reach out or go behind the shelf. What you'll be seeing today is just a prototype made out of cardboard and hence, one of our group members will manually rotate it since there is no motor yet. There is nothing right or wrong here, we are just testing this concept; your honest opinions are the best. This should take about 20 minutes. Can we take notes or record this?

b. **Warm-up questions:**
    i. Where do you keep things you use every day, like dishes, spices or snacks?
    ii. Which items are hardest for you to reach? Why?
    iii. Have you ever felt unsteady or had a close call while reaching up or bending down for something?
c. **Task questions (ask after each task in Section 6):**
    i. How easy or hard was that? (1 = very hard, 5 = very easy)
    ii. Were the buttons easy to see, read and press?
    iii. Did the rotation speed feel safe and comfortable?
d. **Concept survey (1–5 scale, like the electric scooter example):**
    i. How easy does SpinShelf seem to use? (1 = very hard, 5 = very easy)
    ii. How much would you trust a motorized cabinet like this? (1 = not at all, 5 = completely)
    iii. How safe would you feel using it? (1 = not safe, 5 = very safe)
    iv. Compared with your current cabinets, how useful would this be? (1 = not useful, 5 = very useful)
    v. How likely would you be to use one in your home? (1 = very unlikely, 5 = very likely)
    vi. How likely would you be to buy one, or have a family member buy it for you? (1 = very unlikely, 5 = very likely)
e. **Open-ended questions:**
    i. What did you like most about SpinShelf?
    ii. What worried you or felt confusing?
    iii. Is the height right for you? Is the size right for your kitchen?
    iv. Where in your home would you put it, and what would you store in it?
    v. How is this better or worse than a regular lazy susan?
    vi. What would make you trust it more? For example, a stop button or a slower speed.
    vii. What would you expect to pay for it?
    viii. Anything else you'd change or add?

6. Testing Scenarios (if applicable)

#	Scenario	What we're watching for
1	"You're cooking and need the cinnamon. Get it from SpinShelf."	Do they find the right button? Is it obvious? Any hesitation?
2	"Grab your coffee mug from the back section."	Does the rotation bring the item close enough? Any reaching or bending?
3	"Put this jar back where it belongs."	Can they find the right section to return items to?
4	Compare: get a similar item from a high shelf, or a mock-up of one (only if it's safe to do so)	Which one feels easier and safer to them?
5	The teammate rotates slightly slower or faster	Their reaction to speed, and whether faster feels unsafe

7. Roles

Team member	Responsibility
Aadi	Interviewer: runs the intro script, the tasks and the questions; writes the report for sessions 1–2
Minseo Park	"Wizard": rotates the prototype on cue; observer during the interview; writes the report for sessions 3–4
Leo	Notetaker and video/photo recording: logs ratings, quotes and moments of hesitation; writes the report for sessions 5–6